

INTERNATIONAL STANDARD

**Printed board assemblies –
Part 3: Sectional specification – Requirements for through-hole mount soldered
assemblies**



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**Printed board assemblies –
Part 3: Sectional specification – Requirements for through-hole mount soldered
assemblies**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

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PRINTED BOARD ASSEMBLIES –**Part 3: Sectional specification – Requirements
for through-hole mount soldered assemblies**

FOREWORD

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International Standard IEC 61191-3 has been prepared by WG 2: Requirements for electronics assemblies, of IEC technical committee 91: Electronics assembly technology.

This second edition cancels and replaces the first edition published in 1998. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) The requirements have been updated to be compliant with the acceptance criteria in IPC-A-610F.

The text of this standard is based on the following documents:

CDV	Report on voting
91/1375/CDV	91/1435/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61191 series, published under the general title *Printed board assemblies*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

PRINTED BOARD ASSEMBLIES –

Part 3: Sectional specification – Requirements for through-hole mount soldered assemblies

1 Scope

This part of IEC 61191 prescribes requirements for lead and hole solder assemblies. The requirements pertain to those assemblies that totally use through-hole mounting technology (THT), or the THT portions of those assemblies that include other related technologies (i.e. surface mount, chip mounting, terminal mounting).

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60194, *Printed board design, manufacture and assembly – Terms and definitions*

IEC 61191-1:2013, *Printed board assemblies – Part 1: Generic specification – Requirements for soldered electrical and electronic assemblies using surface mount and related assembly technologies*

IPC-A-610, *Acceptability of Electronic Assemblies*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60194 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

THT

through-hole technology

technology that permits an electrical connection of components to a conductive pattern by the use of plated or non-plated holes with annular rings in the mounting substrate

4 General requirements

Requirements of IEC 61191-1 are a mandatory part of this specification. Workmanship shall meet the requirements of IPC-A-610 in accordance with the classification requirements of this document.

5 Through-hole mounting of components

5.1 General

This clause covers the assembly of components with leads inserted into through-holes and soldered by machine and/or manual processes.

5.2 Placement accuracy

Placement accuracy for components inserted either manually or by machine methods shall be sufficient to insure that components are properly positioned after soldering. If suitable process controls are not in place to ensure compliance with this requirement and the intent of Annex A, the detailed requirements of Annex A shall be applicable.

5.3 Through-hole component requirements

5.3.1 Lead preforming

Part and component leads shall be pre-formed to the final configuration, excluding the final clinch or retention bend, before assembly or installation.

5.3.2 Tempered leads

When it is necessary to cut tempered leads, the governing work instructions shall specify cutting tools that do not impart detrimental shocks to internal connections of the components.

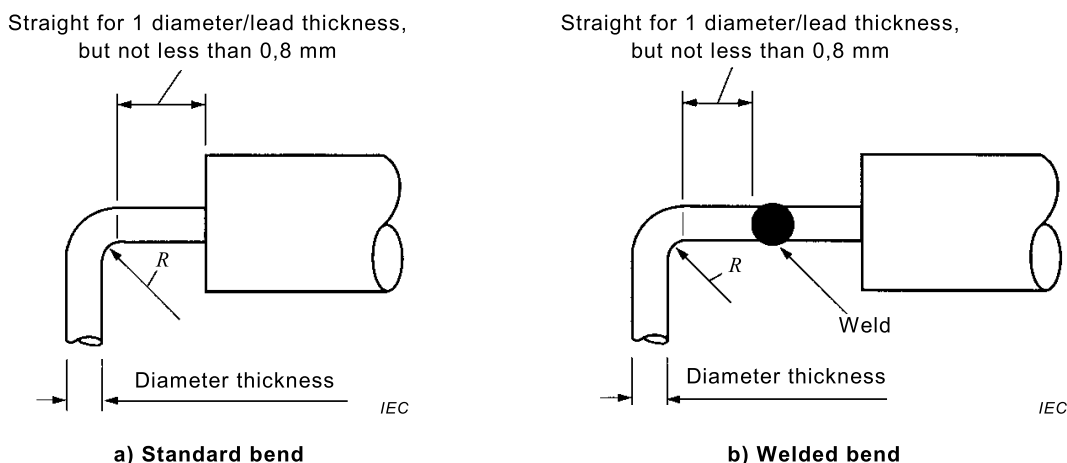
5.3.3 Lead forming requirements

Leads shall be formed in such a manner that the lead-to-body seal is not damaged or degraded. Leads shall extend at least one lead diameter or thickness but not less than 0,8 mm from the body or weld before the start of the bend radius (see Figure 1).

Exposed core metal is acceptable if reduction in the cross-sectional area does not exceed 10 % of the diameter of the lead. Occurrence of exposed core metal in the formed area of the lead shall be treated as a process indicator.

Measurement shall be made from the end of the part. The end of the part is defined to include any coating meniscus, solder seal, solder or weld bead, or any other extension.

The span for components mounted with a conventional lead form is 7,6 mm minimum, and 33 mm maximum.



Maximum lead diameter mm	Minimum bend radius R
Up to 0,8	1 diameter/thickness
0,8 to 1,2	1,5 diameters/thicknesses
Larger than 1,2	2 diameters/thicknesses

Figure 1 – Lead bends

5.3.4 Stress relief requirements

Component leads shall be formed in such a manner that the lead compliancy is not restricted in providing the natural stress relief of the lead material. Special lead formations are permitted to enhance stress relief properties.

5.3.5 Lead termination requirements

5.3.5.1 General

To ensure part retention during soldering operations, lead terminations in printed board plated through-holes shall be one of the following configurations: full clinch, partially clinched, or straight-through lead termination, as specified on the assembly drawing. In the event that nothing is specified, the manufacturer shall conform to the requirements in 5.3.5.2 to 5.3.5.10 as appropriate.

5.3.5.2 Clinched lead termination

Leads in unsupported holes and class C shall be clinched a minimum of 45° . The lead end should not extend beyond the edge of the land; however, if overhang does occur, the lead extension shall not violate minimum electrical spacing requirements. Leads formed of alloy 42 or comparable iron bearing alloys shall not be terminated with a full clinch.

NOTE Alloy 42 has the composition of Fe-Ni 41-Mn 0,8-Co 0,5.

5.3.5.3 Clinched lead orientation

When manually clinched, the clinched portion of the wire or lead should be directed along a printed conductor connected to the land. The leads on opposite ends or sides of a component should be directed in opposite directions. When automatically clinched, the orientation of the clinch relative to any conductor is optional. Manually formed clinches for non-axial leaded components should be directed radially from the centre of the component when the termination area array on the printed board is patterned for such radial orientation.

5.3.5.4 Partially clinched leads

Partially clinched leads shall be bent sufficiently to provide the necessary mechanical restraint during the soldering process. Alternate bend directions may be used. Diagonally opposite corner leads on dual-in-line packages (DIPs) may be partially clinched to retain parts during soldering operations. DIP leads should be bent outwards away from the longitudinal axis of the body.

5.3.5.5 Straight-through lead terminations

Component leads terminated straight through shall not extend in excess of 1,5 mm for level C, or 2,5 mm for level B, and no danger of shorts for level C, beyond the conductor surface, and as a minimum shall be visible in the completed solder connection. The lead protrusion for unsupported holes shall be discernible in the solder joint for levels A and B, and sufficient to clinch for level C. The maximum protrusion in all levels shall not create the risk of shorts. Assembly designs that necessitate different lead extensions are considered to have unique mounting requirements to be noted on the approved assembly drawing.

5.3.5.6 Meniscus spacing and trimming

Components shall be mounted to provide a visible clearance between the coating meniscus on each lead and the solder connection. Trimming of the component coating meniscus is prohibited.

5.3.5.7 Lead trimming

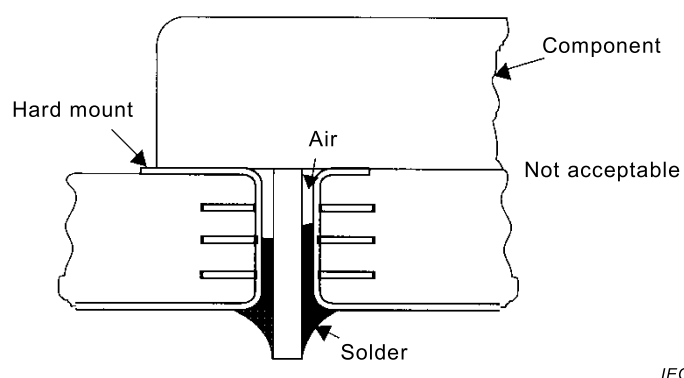
Leads may be trimmed after soldering, provided the cutters do not damage the component or solder connection due to physical shock. When lead cutting is performed after soldering, the solder terminations shall either be reflowed or visually inspected at 10 times magnification to ensure that the original solder connection has not been damaged (e.g. fractured) or deformed. If the solder connection is reflowed, this shall be considered a part of the soldering process and shall not be considered rework. This requirement is not intended to apply to components that are designed so that a portion of the lead is intended to be removed after soldering (e.g. breakaway tie bars).

NOTE This inspection at 10 times magnification is to evaluate the soldered termination for evidence of physical damage or deformation, indicated by fractures that are smaller than the land sizes inspected in accordance with IEC 61191-1, provided that the breakpoint does not expose lead material that is corrodible (e.g. Kovar™¹ [Fe 54/Ni 29/Co 17])

5.3.5.8 Hole obstruction

Components shall be mounted so that they do not obstruct solder flow onto the upper side lands of plated through-holes required to be soldered (see Figure 2).

¹ Kovar is an example of a suitable product available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of this product.



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Figure 2 – Hole obstruction

5.3.5.9 Metal-cased component insulation

Metal-cased components shall be insulated from adjacent electrically conductive elements. Insulation material shall be compatible with the circuit and printed board material.

5.3.5.10 Jumper wires

Jumper wires shall conform to the applicable design specification(s) and shall be documented on the assembly drawing.

6 Acceptance requirements

6.1 General

Materials, processes and procedures described and specified in IEC 61191-1 provide for soldered interconnections that are better than the minimum acceptance requirements in this clause. Processes and their controls should be capable of producing products meeting or exceeding the acceptance criteria for a level C product.

6.2 Control and corrective actions

6.2.1 General

The detailed requirements for acceptance, corrective action limits, control limit determination and general assembly criteria described in IEC 61191-1 are a mandatory part of this standard. In addition, the criteria defined in Subclauses 6.2.2 to 6.3.3 shall meet the requirements for all through-hole assembly and connection acceptability.

6.2.2 Interfacial connections (vias)

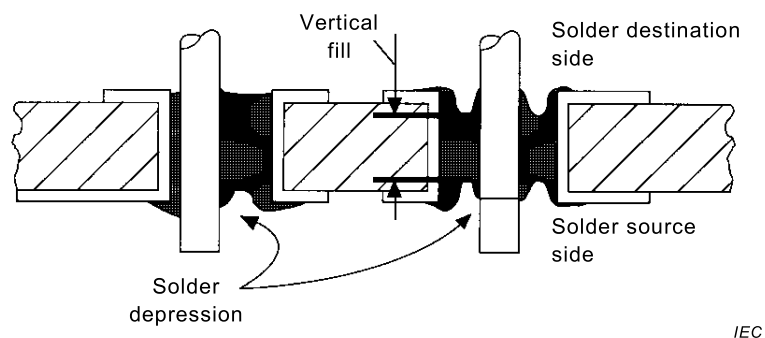
Unsupported holes with leads, or plated through-holes not subjected to mass soldering and used for interfacial connections, need not be filled with solder. Exposed copper on via pads is permitted. Plated through-holes that are not exposed to solder, because of permanent or temporary maskants, and which are used for interfacial connections, need not be filled with solder.

Plated through-holes without leads, including vias, after exposure to reflow, wave, dip, or drag solder processing shall meet the acceptability requirements of IEC 61191-1.

6.3 Through-hole component lead soldering

6.3.1 General

The solder joint shall provide evidence of good wetting, and the plated through-hole solder fill shall meet the requirements of Table 1 and Figures 3 and 5, with solder wetted to the hole's wall. Single-sided boards shall meet conditions C and E of Table 1 and Figure 4.



Minimum acceptable for all levels as in Table 1.

Outline of the lead or wire shall be visible in the solder fillet.

Figure 3 – Through-hole component lead soldering

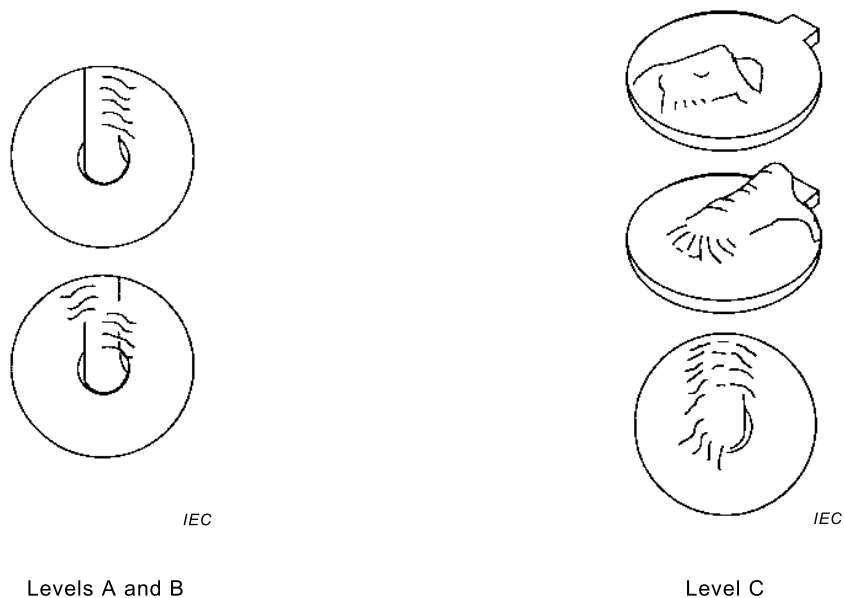


Figure 4 – Lead-to-land fillet requirements for clinched leads and wires in non-plated through-holes

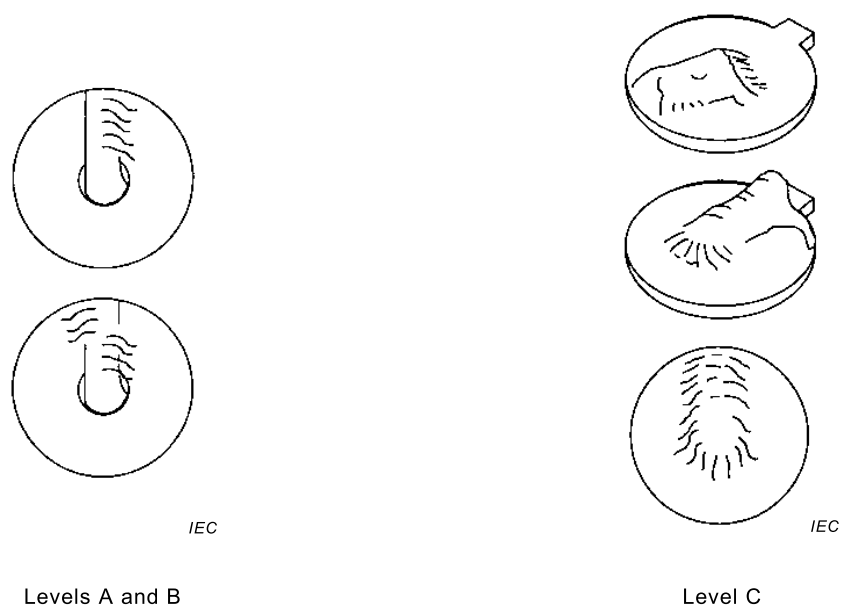


Figure 5 – Lead-to-land fillet requirements for clinched leads and wires in plated through-holes

Table 1 – Plated through-holes with component leads, minimum acceptable conditions¹⁾

Criteria	Level A	Level B	Level C
A. Circumferential wetting-solder, destination side – lead and barrel	Not specified	180°	270°
B. Vertical fill of solder ²⁾	Not specified	75 % ³⁾	75 %
C. Circumferential fillet and wetting – solder source side	270°	270°	330°
D. Percentage of original land area covered with wetted solder – solder destination side	0 %	0 %	0 %
E. Percentage of original land area covered with wetted solder – solder source side	75 %	75 %	75 %
¹⁾ Wetted solder refers to solder applied by the solder process. ²⁾ A total maximum of 25 % depression, including both solder source and destination sides, is permitted. ³⁾ For level B solder fill of 50% or 1.2mm, whichever is less, is acceptable when the plated hole is connected to a heatsink layer, the lead is discernible on the solder source side, and the solder joint shows 360° circumferential wetting of barrel and lead on the solder source side.			

The connection for level A and B may have one or two fillets corresponding in length to 270° of the overlap of the lead on the termination land.

A connection in level C shall show circumferential fillet of 330° complete. Heel of leads shall be wetted with solder. Solder shall fill the hole in accordance with table 1.

6.3.2 Clinched leads

If a lead or wire is clinched, the lead or wire may make contact with the conductor pattern before soldering. The outline of the lead or wire shall be visible in the solder fillet. See Figures 4 and 5.

6.3.3 Exposed basis metal

Incomplete solder wetting at the tip of through-hole component leads, the edges and/or periphery of printed board lands and conductors is acceptable. Exposed basis metal caused by cutting of component leads after soldering is permissible (see 5.3.5.7).

7 Rework of unsatisfactory solder connections

Rework of unsatisfactory solder connections shall not be performed until the discrepancies have been documented. This data shall be used to provide an indication as to possible causes, and to determine if corrective action is required in accordance with IEC 61191-1. When rework is performed, each reworked/or reflowed connection shall be inspected to the requirements of 5.3 (see Table 2 for list of defects).

Table 2 – Through-hole solder joint defects

No.	Defects
1	Defects identified in Table 2 of IEC 61191-1:20XX
2	Solder connections that do not meet the requirements of Table 1 exhibiting a solder fillet joining the component lead to the land
3	Inadequate stress relief on components and wires
NOTE Solder in the stress relief bend does not constitute elimination of stress relief.	

Annex A (normative)

Placement requirements for through-hole mount devices

A.1 General

The following placement requirements for through-hole mount devices shall be imposed only if process controls are not sufficiently in place to ensure compliance with 5.1.

A.2 Horizontal mounting, free-standing

When components are mounted free-standing, the spacing between the body of the component or end of the meniscus (if present) and the surface of the board shall, as a minimum, meet the cleaning requirements of the assembly or the requirements stated in 5.2. As a maximum, the spacing should be 2,0 mm. In no instance shall non-parallelism result in non-conformance with the minimum spacing limits (see Figure A.1).

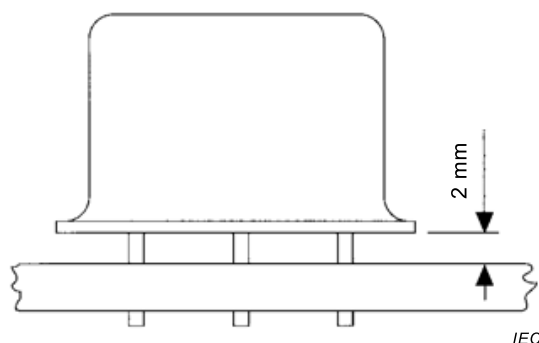


Figure A.1 – Mounting of free-standing components

A.3 Axial lead components

Axial lead parts shall be mounted as specified on the approved assembly drawing and mounted approximately parallel to the board surface, or perpendicular as specified in A.4.1. Bodies of axial lead parts should be approximately centred.

A.4 Radial lead components

Radial lead components should be mounted parallel to the surface of the printed board within the spacing tolerances specified herein.

A.5 Perpendicular mounting, free-standing

A.5.1 General

The spacing requirements for free-standing, perpendicular mounted parts shall be the same as those for horizontal mounting (see Clause A.1).

A.5.2 Mounting of components

Unless otherwise noted on the assembly drawing, through-hole components required to be perpendicularly mounted should be installed with their major axis approximately 90° to the

board surface and with a minimum of 0,3 mm between the component body (seal or lead weld) and the board surface.

A.5.3 Radial lead components

When dual-lead components are mounted free-standing, the larger sides should be within 15° of perpendicular to the board surface. This requirement applies to components of configurations shown in Figures A.2 and A.3.

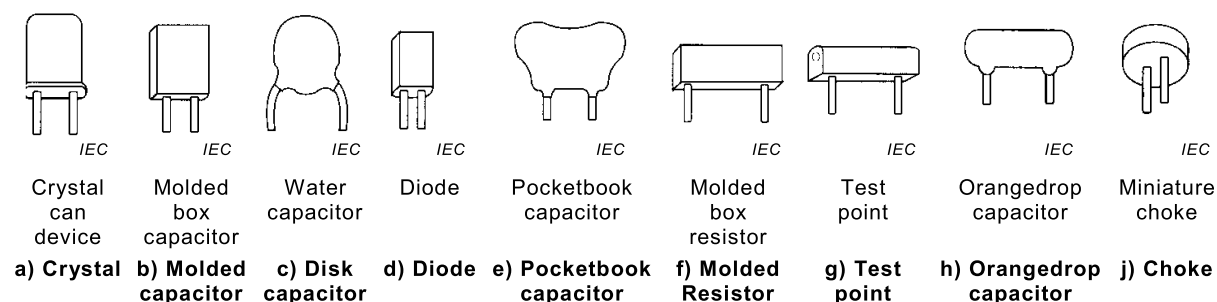


Figure A.2 – Typical configuration of components with dual non-axial leads

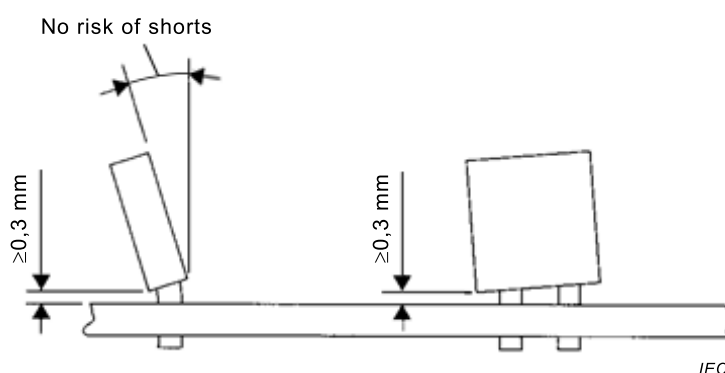


Figure A.3 – Mounting of components with dual non-axial leads

A.6 Side- and end-mounting

When documented on an approved assembly drawing, a component may be either side-mounted or end-mounted as shown in Figures A.4 and A.5. The side or surface of the body, or at least one point of any irregularly configured component (such as certain pocketbook capacitors) shall be in full contact with the printed board, and the body shall be bonded or otherwise retained to the board to prevent damage when vibrational and shock forces are applied.

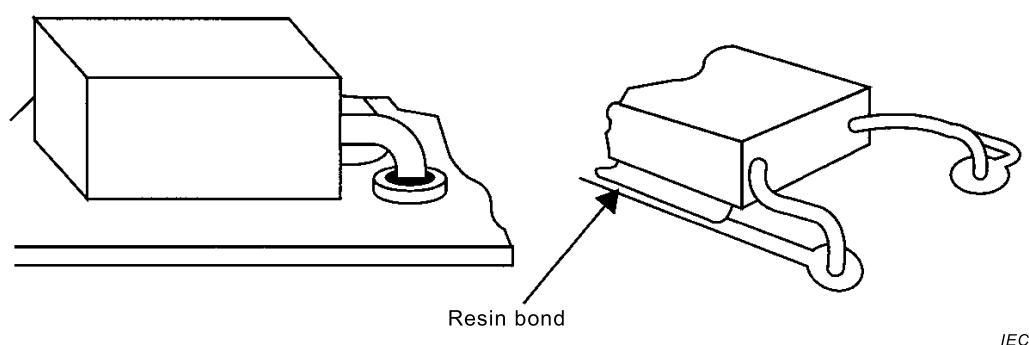


Figure A.4 – Side mounting

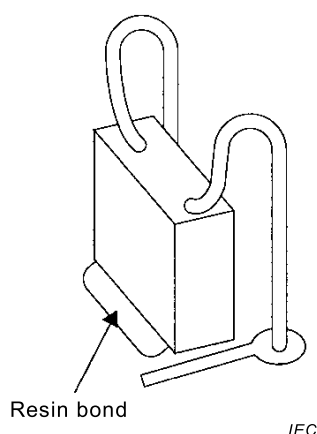


Figure A.5 – End mounting

A.7 Supported component mounting

A.7.1 General

When components are supported, they shall be mounted as follows.

When a component with resilient integral feet or a resilient integral stand-off is mounted to a printed board, the component shall be seated with each foot in contact with the surface of the board. For this requirement, a button stand-off as shown in Figure A.6 b) shall be deemed a foot and the mating surface of each button shall be flat on the bare board or circuitry.

- a) resilient feet or stand-offs integral to the component body (see Figures A.6 a) and A.6 b));
- b) resilient or specially configured non-resilient stand-off devices (see Figure A.6 c)); or
- c) separate resilient, non-footed stand-offs which do not block plated through holes nor conceal connections on the component side of the board.

When a component with resilient integral feet or a resilient integral stand-off is mounted to a printed board, the component shall be seated with each foot in contact with the surface of the board. For this requirement, a button stand-off as shown in Figure A.6 b) shall be deemed a foot and the mating surface of each button shall be flat on the bare board or circuitry.

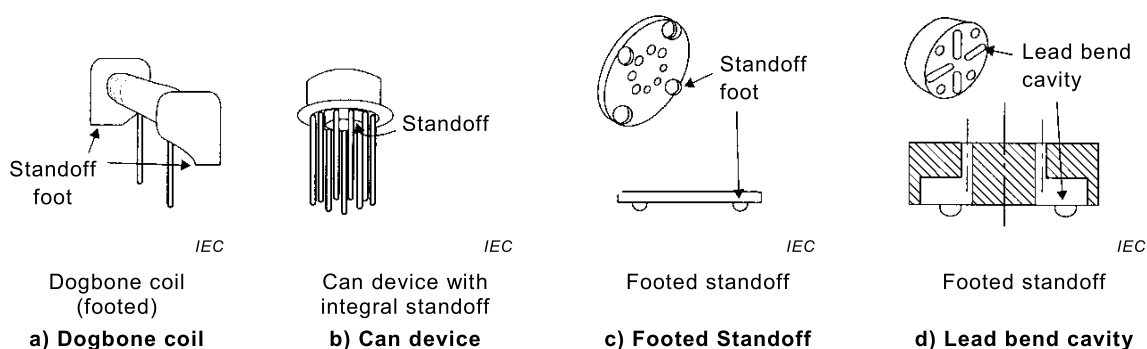


Figure A.6 – Mounting with footed stand-offs

A.7.2 Stand-off positioning

No stand-off shall be inverted.

A.7.3 Non-resilient footed stand-offs

When specially configured non-resilient stand-offs are used, that portion of the lead in the lead bend cavity (see Figure A.7 b)) shall conform with an angular line extending from the lead insertion hole in the stand-off device to the land attachment hole in the printed board.

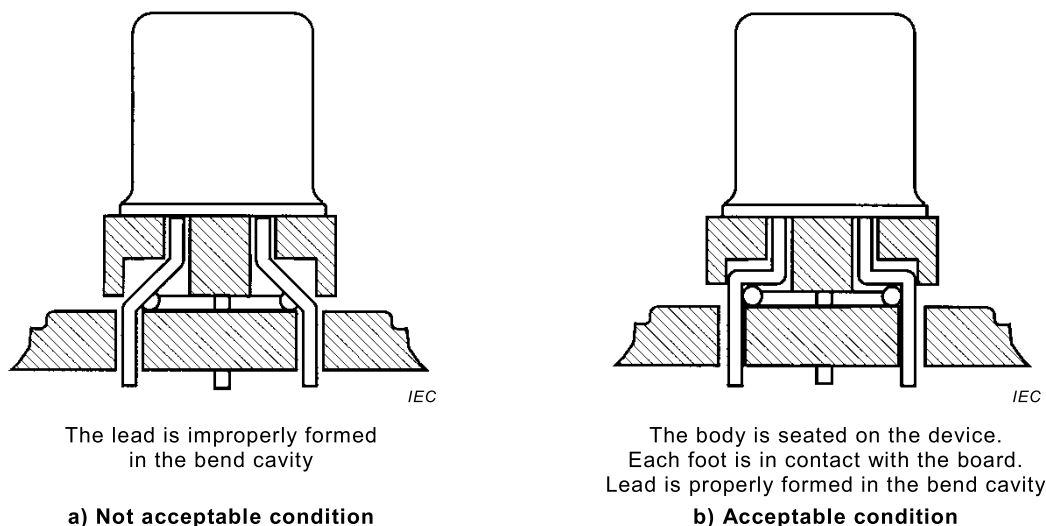


Figure A.7 – Non-resilient footed stand-offs

A.8 Stress relief lead configuration

Components shall be mounted to any one, or a combination of, the following configurations:

- in a conventional manner using 90° (nominal) lead bends directly to the mounting hole (see Figure A.8 a));
- with camel hump bends (see Figures A.8 b) and A.8 c)). Configuration incorporating a single camel hump (see Figure A.8 b)) may have the body positioned off-centre;
- other configurations may be used with an agreement of the user.

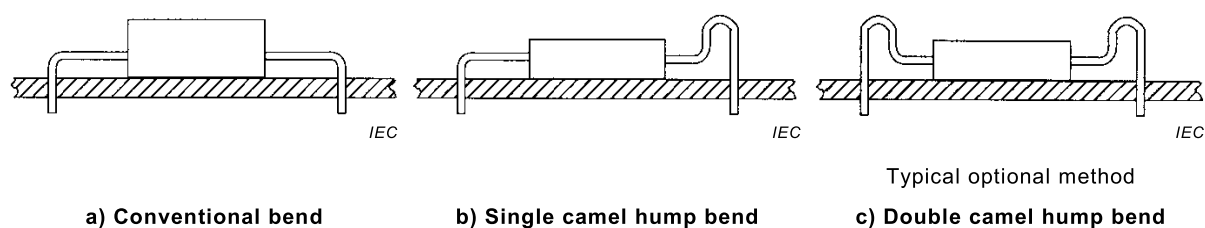
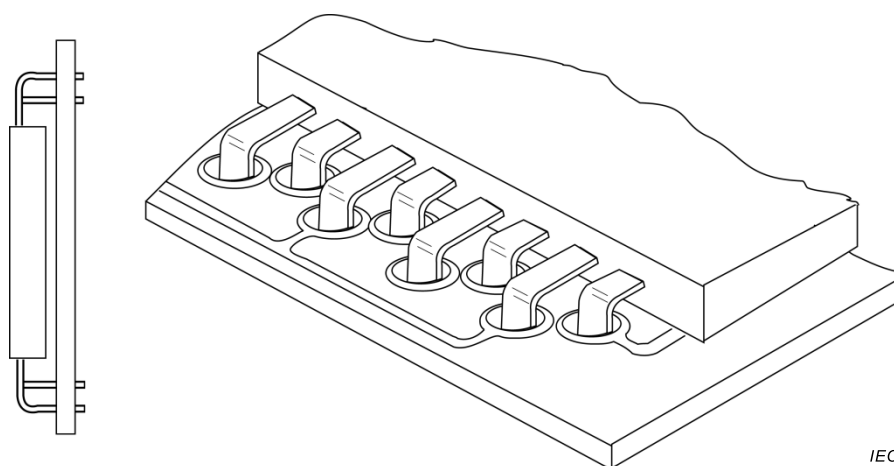


Figure A.8 – Acceptable lead configurations

A.9 Flat pack lead configuration

Whether it be ribbon, flattened, or square configuration, leads of flatpacks of the normal and butterfly configurations (leads leaving from two or more side walls) can be configured as shown in Figure A.9.



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Figure A.9 – Configuration of ribbon leads for through-hole mounting

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IEC 60068-2-58, *Environmental testing – Part 2-58: Tests – Test Td: Test methods for solderability, resistance to dissolution of metallization and to soldering heat of surface mounting devices (SMD)*

IEC 61188-5-1, *Printed boards and printed board assemblies – Design and use – Part 5-1: Attachment (land/joint) considerations – Generic requirements*

IEC 61188-5-2, *Printed boards and printed board assemblies – Design and use – Part 5-2: Attachment (land/joint) considerations – Discrete components*

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